

Nadeem Mohamed Soliman

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PROFILE

Senior Computer Science and Engineering student at the German University in Cairo, graduating June 2026 with an **A+ Bachelor's degree**. I have shipped backend systems, data pipelines, and AI applications across banking, logistics, and enterprise software. I focus on building services that hold up under load, pipelines that don't need babysitting, and code that the next engineer can actually understand.

EDUCATION

German University in Cairo (GUC)

Cairo, Egypt

Bachelor of Science in Computer Engineering and Science

Sep 2021 – Jun 2026 (Expected)

- **Cumulative GPA: 1.24 / 5.0** (Top of class; German grading scale where 1.0 is the highest possible grade)
- Bachelor Thesis: Grade **A+** in Vision Transformers for Autonomous Vehicle Intention Prediction
- Relevant Coursework: Cloud Computing, Data Structures and Algorithms, Operating Systems, Computer Networks, Database Systems, Network Security, Project Management

BACHELOR THESIS

Vision Transformer Framework for Vehicle Intention Prediction | *PyTorch, Python*

Feb 2025 – Jun 2025

- Trained a Vision Transformer (ViT) to predict vehicle maneuvers in autonomous driving scenarios, beating Uber's baseline LSTM model by **52.9%**.
- Worked on the **Argoverse 2** dataset, handling data preprocessing and temporal feature extraction as part of the end-to-end training pipeline.

EXPERIENCE

Software Engineer | Turuq

Oct 2025 – Present

- Built an embedded Shopify app in **Remix, React, and Node.js** that connects merchant stores to a logistics backend via Shopify Webhooks and GraphQL, handling real-time order events across multiple accounts.
- Shipped a bulk waybill generation feature that automated the full shipment creation workflow, **cutting manual processing time by 80%** across all merchant accounts.
- Contributed to a courier management service on **Bun, Hono, Clerk, MongoDB, and Next.js** for real-time order tracking and courier dispatch.
- Built a customer communication service on top of the **Meta Cloud API**, handling inbound and outbound messaging, conversation state across multi-step flows, and automated order confirmations via HMAC-verified webhooks.
- Consolidated WhatsApp, SMS, and email into a single modular **TypeScript** backend service using inheritance and polymorphism, so new channels can be added without touching existing integrations.

Backend Developer Intern | Ejada

Aug 2025

- Built **Spring Boot microservices** for a virtual banking platform, wiring in **Apache Kafka** for async inter-service messaging and **WSO2 API Gateway** for authenticated, rate-limited external routing.
- Delivered REST APIs for mobile and web banking clients with **OAuth 2.0** authentication, role-based access control, and request validation across all endpoints.

Backend Developer Intern | Link Development

Jul 2025 – Aug 2025

- Integrated **Keycloak** SSO into **OpenKM**, replacing a fragmented login system and consolidating identity management inside a Dockerized MSSQL environment.
- Built automated backup and recovery pipelines using **Rclone** to meet compliance and disaster recovery requirements.
- Wrote deployment and onboarding documentation that became the team's standard reference, **cutting engineer ramp-up time by 60%**.

System Development Intern | Electrolux

Aug 2024 – Sep 2024

- Built an automated quality tracking system in **VBA** (Excel and Access) for the production floor, replacing a fully manual process and **cutting data entry time by 70%** while improving reporting accuracy.

Full Stack Intern | QNB Egypt

Jul 2024

- Built a customer survey platform in **Java**, **XHTML**, and **MySQL** that let branch teams collect, store, and review structured feedback.

Technical Intern | BVS Company

Jul 2024 – Sep 2024

- Completed structured training in **IBM DataPower** API Gateway configuration, ACH payment processing, and enterprise banking security protocols.

Software Development Intern | The Sparks Foundation

Jul 2023

- Delivered a full-stack **MERN** banking app with user authentication, account management, and transaction history, end to end in one month.

UNIVERSITY PROJECTS

Massively Scalable Food Delivery Platform (Talabat-like) | *Spring Boot, PostgreSQL, MongoDB, Redis, JWT*

- Contributed to a distributed food delivery platform where each microservice owned its own database and communicated through well-defined APIs, connecting customers, restaurants, and delivery drivers.
- Owned the **User Service** end to end, implementing JWT-based authentication with **BCrypt password hashing** and role-based access control for CUSTOMER and ADMIN roles.
- Applied the **Observer design pattern** to wire in MongoDB audit logging without touching the business layer, so every user state change was recorded automatically.
- Added a **Redis caching layer** with TTL expiration and cache invalidation on writes to cut unnecessary database calls and keep response times low under load.

Agentic NDA Contract Analysis System | *LLMs, QLoRA, Agentic AI, NLP*

- Fine-tuned a **Qwen** language model using **QLoRA** on the **ContractNLI benchmark**, training it to classify 17 legal hypotheses per contract as ENTAILED, CONTRADICTED, or NOT MENTIONED.
- Built the **vector embedding layer** for semantic search over contract clauses, and helped design the system architecture across reasoning, policy, orchestration, and logging components.
- Worked in a 5-member team building an agentic contract review pipeline; my focus was output reliability and consistency across real-world legal documents.

Athletic Tournament Management System | *IT Project Management*

- Contributed to the **planning and initiation phase** of a digital tournament management system for the German University in Cairo, covering both students and staff.
- Delivered a **Project Charter**, a formal **Scope Statement**, and a **Work Breakdown Structure** with 3 levels of hierarchy and 10+ sub-activities, following standard project management practices.
- Mapped stakeholder interests and set up a **Jira board simulation** that translated WBS elements into epics, tasks, and subtasks with assigned owners and dependencies.

Real-Time Stock Analytics Platform | *Kafka, Spark, Airflow, Docker*

- Built an end-to-end data pipeline using Apache Airflow for orchestration and Kafka with PySpark for real-time market data streaming.
- Stored processed data in PostgreSQL, integrated a GenAI-powered Text-to-SQL agent using LangChain, and built interactive dashboards with Apache Superset.

Graph-RAG Travel Assistant | *Python, Neo4j, LangChain*

- Built a knowledge graph from over **50,000 hotel reviews**, implemented Cypher-based retrieval with named entity recognition, and deployed the system as a Streamlit app.
- Designed the retrieval layer to combine vector similarity search with graph traversal, so LLM answers were grounded in structured, verifiable data. Added SHAP and LIME explainability to the ML models.

All-in-One Vacation Platform | *MERN, Stripe API, Amadeus API*

- Built a full-stack travel booking platform where users can search and book flights and hotels, apply promo codes, and get personalized recommendations.
- Integrated Stripe for payments and Amadeus for live booking data, and worked in an Agile team using Git and GitHub.

Embedded Auto Lane Detection | *C, Hardware Integration*

- Built a real-time lane detection system in C that interfaced directly with hardware drivers to support autonomous navigation on an embedded platform.

University Academic System and Portal | *SQL Server, JavaScript, HTML, CSS*

- Designed an academic management system with a normalized relational schema and 50+ stored procedures, views, triggers, and functions to automate registration, graduation checks, and probation rules.
- Built a role-based web portal for students, advisors, and administrators.

Donation Ecosystem Platform | *React.js*

- Built a platform connecting donors with NGOs, took part in requirements gathering, and delivered a responsive, accessible React prototype.

The Last of Us: Legacy Game Engine | *Java OOP*

- Built a turn-based survival game in Java using OOP principles and MVC architecture, with a custom GUI, a grid-based map, and thorough exception handling.

VOLUNTEERING AND LEADERSHIP

Director, Juniors Technical Program | IEEE GUC Student Branch

Sep 2024 – Jul 2025

- Led the technical program for the branch, coordinating a team to plan and run **36 technical sessions** across the academic year, with close to **50 students** learning from our team.

Junior Teaching Assistant | GUC, Introduction to Programming (CSEN 102)

Sep 2023 – Jan 2024

- Ran weekly Python lab sessions for **20+ students**, covering data structures, algorithms, and object-oriented programming.

CERTIFICATIONS

Data Structures and Algorithms (Microsoft): Algorithmic problem solving and optimization. *(November 2025)*

Web Development (Microsoft and Sprints): Python OOP, RESTful API design, HTML5, CSS3. *(October 2025)*

Machine Learning and AI (Microsoft and Sprints): End-to-end ML pipelines, EDA, Pandas, Scikit-learn, and model deployment. *(September 2025)*

DevOps (Microsoft and Sprints): SDLC, Docker, CI/CD, and AWS (EC2, S3, RDS). *(August 2025)*

Google Cloud Fundamentals: Core Infrastructure (Google Cloud): Core GCP services, IAM, networking, and resource hierarchy.

Essential Google Cloud Infrastructure: Foundation (Google Cloud): Compute Engine, storage services, and deployment basics.

Essential Google Cloud Infrastructure: Core Services (Google Cloud): Load balancing, autoscaling, monitoring, and cloud operations.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, TensorFlow (Familiar), Scikit-learn, Computer Vision, NLP, Model Training & Optimization, Hyperparameter Tuning (Optuna), Vision Transformers (ViT)

Data Engineering: Apache Spark (PySpark), Apache Kafka, Apache Airflow, ETL Pipelines, Stream Processing, Data Warehousing, Apache Superset

Databases & Graph: PostgreSQL, MySQL, MongoDB, MSSQL, Redis, Neo4j, SQL Optimization, B+ Tree Indexing, Knowledge Graphs

Backend & APIs: Java, Spring Boot, Microservices, Node.js, Express.js, REST APIs, GraphQL, OAuth 2.0, Message Queues, WSO2, Event-Driven Architecture

Cloud & DevOps: Docker, Kubernetes (Familiar), Git, GitHub, Linux, GCP, AWS (EC2, S3, RDS), CI/CD Pipelines, IBM DataPower

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML, CSS

Libraries & Frameworks: Pandas, NumPy, LangChain, Streamlit, React, Remix, SHAP, LIME

Concepts: Distributed Systems, System Design, Design Patterns, RAG Systems, OOP, Data Structures, Algorithms, Networking, Agile/Scrum